MorphGrower: A Synchronized Layer-by-layer Growing Approach for Plausible Neuronal Morphology Generation

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Main result: Generated trees look real and remain topologically valid

- Multi-view real/fake test: near-chance accuracy (about 55–63%) indicates high visual plausibility, vs. MorphVAE detected at ~81–95%.
- Topological validity: 100% by enforcing binary branching during growth.
- Evaluated on four datasets used for benchmarking: VPM (ventral posteromedial nucleus), RGC (retinal ganglion cells), M1-EXC (motor cortex excitatory), M1-INH (motor cortex inhibitory).

Table: Classification accuracy (%)

Dataset/Method	VPM	RGC	M1-EXC	M1-INH
MorphVAE	86.75 ± 06.87	94.60 ± 01.02	80.72 ± 10.58	91.76 ± 12.14
MorphGrower	54.73 ± 03.36	62.70 ± 05.24	55.00 ± 02.54	54.74 ± 01.63
MorphGrower†	59.47 ± 04.53	54.99 ± 03.08	55.26 ± 02.42	55.47 ± 03.43

Key quantitative results across four datasets

- Closer match to real neurons on six standard morphology statistics.
- More realistic branching orientations (angles) than MorphVAE.
- Strong performance even without provided soma branches (†).

Metric acronyms: MBPL = Mean Branch Path Length; MMED = Mean Euclidean Distance; MMPD = Mean Path Distance; MCTT = Mean Contraction; MASB = Mean Angle between Sibling Branches; MAPS = Mean Angle between Parent and Sibling.

Table: Performance on the four datasets by the six quantitative metrics

Dataset	Method	MBPL / μm	MMED / μm	MMPD / μm	MCTT / %	MASB / °	MAPS / °
	Reference	51.33 ± 0.59	162.99 ± 2.25	189.46 ± 3.81	0.936 ± 0.001	65.35 ± 0.55	36.04 ± 0.38
M	MorphVAE	41.87 ± 0.66	126.73 ± 2.54	132.50 ± 2.61	0.987 ± 0.001		
VP	MorphGrower	48.29 ± 0.34	161.65 ± 1.68	180.53 ± 2.70	0.920 ± 0.004	72.71 ± 1.50	43.80 ± 0.98
	MorphGrower†	46.86 ± 0.57	159.62 ± 3.19	179.44 ± 5.23	0.929 ± 0.006	59.15 ± 3.25	46.59 ± 3.24
	Reference	26.52 ± 0.75	308.85 ± 8.12	404.73 ± 12.05	0.937 ± 0.003	84.08 ± 0.28	50.60 ± 0.13
RGC	MorphVAE	43.23 ± 1.06	248.62 ± 9.05	269.92 ± 10.25	0.984 ± 0.004		
	MorphGrower	25.15 ± 0.71	306.83 ± 7.76	384.34 ± 11.85	0.945 ± 0.003	82.68 ± 0.53	51.33 ± 0.31
	MorphGrower†	23.32 ± 0.52	287.09 ± 5.88	358.31 ± 8.54	0.926 ± 0.004	76.27 ± 0.86	49.67 ± 0.41
	Reference	62.74 ± 1.73	414.39 ± 6.16	497.43 ± 12.42	0.891 ± 0.004	76.34 ± 0.63	46.74 ± 0.85
M1-EXC	MorphVAE	52.13 ± 1.30	195.49 ± 9.91	220.72 ± 12.96	0.955 ± 0.005		
	MorphGrower	58.16 ± 1.26	413.78 ± 14.73	473.25 ± 19.37	0.922 ± 0.002	73.12 ± 2.17	48.16 ± 1.00
	MorphGrower†	54.63 ± 1.07	398.85 ± 18.84	463.24 ± 22.61	0.908 ± 0.003	63.54 ± 2.02	48.77 ± 0.87
	Reference	45.03 ± 1.04	396.73 ± 15.89	705.28 ± 34.02	0.877 ± 0.002	84.40 ± 0.68	55.23 ± 0.78

Electrophysiology: waveforms align with real neurons

- Average membrane potential (MP) traces of generated vs. real neurons closely overlap.
- Small relative errors on summary electrophysiology metrics (e.g., mean frequency and after-potential measures).
- Validates plausibility from a biophysical standpoint using the NEURON (simulation environment).

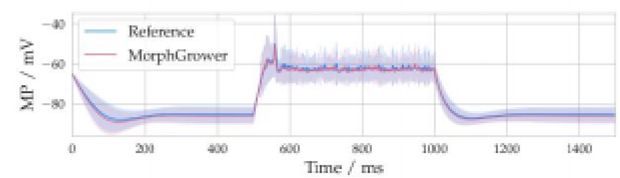

Figure: Average electrophysiology traces: real vs. generated (MP: membrane potential).

Method in one slide: synchronized, layer-wise growth

- Generate neurons layer-by-layer from the soma; decode sibling branches synchronously at each bifurcation.
- Conditional Variational Autoencoder (VAE) with a von Mises–Fisher (vMF) latent space drives robust sampling.
- Conditions on the already-grown structure for plausibility at each step.

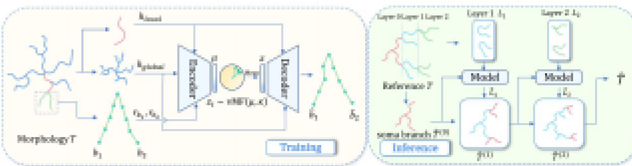

Figure: Architecture overview: encode branch-pairs, sample vMF(μ, κ), and decode sibling branches layer-wise.

Global + Local context (GNN/GRU and EMA) for realistic growth

- Global context: treat prior layers as a forest and aggregate with a Graph Neural Network (GNN) using a Gated Recurrent Unit (GRU) bottom-up pass.
- Local context: ancestor-path Exponential Moving Average (EMA) emphasizes recent ancestry for directional continuity.
- Together, these contexts capture inter-branch dependencies and self-avoidance.

Synchronized growth ensures validity and realism

- Generate all sibling pairs in parallel within a layer, conditioned on prior layers.
- Enforces binary branching and avoids topological violations by design.
- Enables fine-grained, development-like growth.

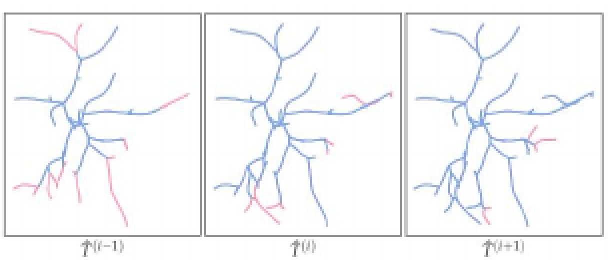

Figure: Intermediate growth visualization: new layer (pink) conditioned on previous morphology (blue).

Evaluation: datasets and tests

- Datasets (evaluation only): VPM, RGC, M1-EXC, M1-INH.
- Tests: morphology statistics, visual plausibility (multi-view classification), Sholl analysis, and electrophysiology (NEURON).
- No dataset collection details are shown; focus is on outcomes.

Conclusion

- Biology-inspired, synchronized layer-wise growth enforces topology and realism.
- Conditional VAE + global/local context (GNN/GRU, EMA) captures key growth dependencies.
- Consistently outperforms MorphVAE on morphology statistics, plausibility, and electrophysiology across datasets.
- Ablations (not shown): both global and local contexts matter.

Questions & discussion

- Thank you!
- Questions and feedback are welcome.